

Emploid Trust Score Algorithm

1. Overview

The Listing Trust Score (stored in the database as `ghost\_score`) is a core feature of the Emploid platform. It uses a multi-factor weighting algorithm to estimate the probability that a job posting represents a real, active hiring opportunity, rather than a stagnant, evergreen, or "ghost" listing.

The score ranges from **0 (extremely likely to be fake/inactive)** to **100 (verified active and real)**.

2. Factor Breakdown & Weights

The score is computed as a weighted sum of **8 individual factors**, each assessing a distinct aspect of the listing and company data.

Factor	Weight	Evaluation Focus
1. Days Active	20%	Time elapsed since the job was posted or first discovered.
2. Repost Frequency	15%	The number of times the listing has been taken down and re-advertised.
3. Company Hiring Velocity	15%	The ratio of active openings to total company headcount.
4. Salary Disclosed	10%	Transparency of pay ranges and whether they are verified or estimated.
5. Hiring Contact Identified	10%	Presence of a linked recruiter, hiring manager, or contact details.
6. Application Method	10%	Direct links to corporate Applicant Tracking Systems (ATS) vs. third-party easy-applies.
7. Description Specificity	10%	Contextual quality indicators within the job description text.
8. Interview Review Recency	10%	External user reports indicating active interviewing at the company.

3. Detailed Logic & Calculations

Factor 1: Days Active (20% Weight)

Objective: Stale listings are likely filled or abandoned.

- **Calculation:** Measures days since `posted\_at` (or `first\_seen\_at` if missing).

$$\text{Days} = \text{Current Date} - \text{Posted Date}$$

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Scoring Logic:

- ≤ 14 days: **100 points** (Highly likely to be active).
- > 90 days: **0 points** (High probability of being a stagnant listing).
- Between 15 and 90 days: Score decays linearly from 100 to 0:

$$\text{Score} = 100 - \left\lfloor \left(\frac{\text{Days} - 14}{76} \right) \times 100 \right\rfloor$$

Factor 2: Repost Frequency (15% Weight)

Objective: Continual reposting indicates evergreen "resume collection" roles rather than active immediate hires.

- **Calculation:** Relies on the `repost\_count` incremented during scraper runs.
- **Scoring Logic:**
- 0 reposts: **100 points** (Appears to be a unique, authentic opening).
- ≥ 4 reposts: **0 points** (Evergreen listing indicator).
- Between 1 and 3 reposts: Score decreases by 25 points per repost:

$$\text{Score} = 100 - (\text{Reposts} \times 25)$$

Factor 3: Company Hiring Velocity (15% Weight)

Objective: A small company posting dozens of listings is a red flag.

- **Calculation:** Compares the company's active openings count (`total\_active\_listings`) with its size bucket (`size\_range`).
- Size Ranges mapped to headcount estimates:
 - `1-50` $\rightarrow$ 50 employees
 - `51-200` $\rightarrow$ 200 employees
 - `201-500` $\rightarrow$ 500 employees
 - `501-1000` $\rightarrow$ 1000 employees
 - `1001-5000` $\rightarrow$ 5000 employees
 - `5000+` $\rightarrow$ 10000 employees
- Velocity Ratio calculation:

$$\text{Ratio} = \frac{\text{Active Listings}}{\text{Estimated Size}}$$

- **Scoring Logic:**
- $\text{Ratio} > 0.10$ ($> 10\%$ of company size open): **0 points** (Suspicious velocity).
- Ratio between 0.05 and 0.10 ($5\% - 10\%$): **50 points** (Moderate hiring).
- $\text{Ratio} \leq 0.05$ ($\leq 5\%$): **100 points** (Reasonable hiring activity).
- *Fallback (no company size data available):* **50 points**.

Factor 4: Salary Disclosed (10% Weight)

Objective: Serious employers disclose pay ranges; fake/ghost listings tend to omit details.

- **Calculation:** Assesses ``salary_min``, ``salary_max``, and the boolean flag ``salary_is_estimate``.
- **Scoring Logic:**
- Salary is disclosed directly by the employer (``salary_is_estimate` = false`): **100 points**.
- Salary is estimated by a third-party aggregator (``salary_is_estimate` = true`): **60 points**.
- No salary information provided: **0 points**.

Factor 5: Hiring Contact Identified (10% Weight)

Objective: Legitimate jobs are typically owned by a visible hiring contact.

- **Calculation:** Checks for entries in the ``job_recruiters`` join table linked to the job.
- **Scoring Logic:**
- At least 1 recruiter or hiring manager is linked: **100 points**.
- No specific contact is associated: **0 points**.

Factor 6: Application Method (10% Weight)

Objective: "Easy Apply" features are flooded with noise; companies often ignore them. Direct application is higher trust.

- **Calculation:** Analyzes the ``source`` field and the syntax of ``apply_url``.
- **Scoring Logic:**
- Source is ``company_direct`` OR the ``apply_url`` does **not** contain string keywords like ``"easy_apply"`` or ``"easyApply"``: **100 points** (Links directly to corporate careers site/ATS).
- Uses generic easy-apply boards: **0 points** (Candidate applications may never reach a hiring manager).

Factor 7: Description Specificity (10% Weight)

Objective: Vague, short, or generic text indicates a low-effort template or ghost job.

- **Calculation:** Scans description text for 4 key specificity indicators:
 1. **Word Count:** Text contains more than 200 words.
 2. **Tech Stack/Tools:** Mentions specific software/tools (e.g., ``react``, ``nodejs``, ``python``, ``sql``, ``aws``, ``docker``, etc.).
 3. **Team Context:** Mentions team structure (e.g., ``"team of"``, ``"reporting to"``, ``"you will join"``, ``"squad"``).
 4. **Specific Duties:** Mentions duties (e.g., ``"responsibilities"``, ``"what you'll do"``, ``"day-to-day"``).
- **Scoring Logic:**
- 3 indicators present: **100 points** (High-quality, detailed description).
- 0 indicators present: **0 points** (Vague description).
- 1 or 2 indicators present: **33 points per indicator** (33 or 66 points).

Factor 8: Interview Review Recency (10% Weight)

Objective: Candidate reviews confirm active recruitment interviews.

- **Calculation:** Checks external data integration for recent interview review submissions.
 - **Scoring Logic:**
 - At least 1 review found: **100 points** (Company actively interviewing).
 - No recent reviews found: **50 points**.
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4. Final Aggregation & Labels

Once the 8 factor scores (F_1 to F_8) are evaluated, the final Listing Trust Score is computed:

$$\text{Score} = \text{round}(F_1 \times 0.20 + F_2 \times 0.15 + F_3 \times 0.15 + F_4 \times 0.10 + F_5 \times 0.10 + F_6 \times 0.10 + F_7 \times 0.10 + F_8 \times 0.10)$$

Based on the aggregated score, jobs are categorized into one of three Trust Labels (stored as `ghost_label` in the database):

1. **Verified (Score ≥ 80):** High-confidence active roles. Clean metadata, direct link, recent posting.
2. **Uncertain (Score 50 to 79):** Requires candidate caution. May have stale dates or missing salary details.
3. **Likely Ghost (Score < 50):** High probability of being inactive. Evergreen listings, easy-apply loops, or long-active dates.